

Program : Diploma in Biomedical Engineering	
Course Code : 5248	Course Title: Biomedical Instrumentation Lab I
Semester : 5	Credits: 1.5
Course Category : Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester : 45

Course Objectives:

- To provide hands-on experience for the students to impart the circuit level knowledge of various biomedical equipment using for diagnosis and analysis.
- To enable the students to pick up body signals using standard equipment.
- To equip student to operate analytical instruments used in clinical labs.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Electronic circuits		Electronic devices and Circuits	3
ECG, EEG, EMG, BP machine, Hb meter, pH meter		Medical Instrumentation	3
Integrated circuits and other linear circuits		Linear Integrated Circuits	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Implement circuit which simulate ECG, detect QRS and ECG telemetry system.	9	Applying
CO2	Acquire and identify body signals using ECG, EEG, EMG recorders and multiparameter monitor.	15	Applying
CO3	Conduct experiments using colorimeter, pH meter and Hb meter	9	Applying
CO4	Measure blood pressure and conduct audiometry test using suitable machines.	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3			3			
CO3	3			3			
CO4	3	2	2	3		2	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Implement circuit which simulate ECG, detect QRS and ECG telemetry system.		
M1.01	Set up the ECG Simulator circuit and generate the output waveform	3	Applying
M1.02	Implement QRS Detector and generate output waveform	3	Applying
M1.03	Implement modulation and Demodulation circuits of ECG biotelemetry using IC 4046	3	Applying
CO2	Acquire and identify body signals using ECG, EEG, EMG recorders and multiparameter monitor.		
M2.01	Acquire ECG signal in bipolar lead configuration using ECG machine.	3	Applying
M2.02	Plot ECG in 12-lead system using ECG simulator.	3	Applying
M2.03	Demonstrate EEG machine and 10-20 electrode placement in a subject	3	Applying
M2.04	Acquire EMG signal and observe muscle fatigue using EMG machine.	3	Applying
M2.05	Monitor and document body parameters using parameter monitor	1.5	Applying
	Series Test - I	1.5	
CO3	Conduct experiments using colorimeter, pH meter and Hb meter		
M3.01	Conduct experiment to verify Beer Lambert's Law using colorimeter	3	Applying
M3.02	Calibrate pH meter using standard solution	3	Applying

M3.03	Demonstrate of Hb meter	1.5	Applying
CO4	Measure blood pressure and conduct audiometry test using suitable machines.		
M4.01	Measure blood pressure using sphygmomanometer	1.5	Applying
M4.02	Measure blood pressure automatic BP apparatus and compare with sphygmomanometer reading.	1.5	Applying
M4.03	Conduct audiometry test using audiometer and plot graph.	3	Applying
M4.04	Open Ended Experiments**	6	Applying
	Lab Exam - II	1.5	

**** - Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open ended experiments as a group of 4-5. There is no duplication in experiments between groups. Students should prepare the user's manual using suitable software. The circuits may be simulated using suitable software.)

1. Prepare the following documents
 - User's manual for pH meter
 - User's manual for Audiometer
 - User's manual for colorimeter
2. Design the circuit and simulate using suitable software
 - Colorimeter
 - 3- Lead ECG simulator
 - ECG telemetry system

Text / Reference:

T/R	Book Title/Author
T1	Op-Amps and Linear Integrated Circuits”- Ramakant A Gayakwad - PHI- 4th Edition R2 “Handbook of Biomedical Instrumentation”- RS Khandpur- Tata McGraw-Hill Pub., 2006
T2	“Handbook of Biomedical Instrumentation”- RS Khandpur- Tata McGraw-Hill Pub., 2006
R1	Handbook of Analytical Instruments – R S Khandpur , Second edition, McGraw Hill Ltd

Online Resources:

Sl.No	Website Link
1	https://www.youtube.com/watch?v=ALfVjWQVvbY
2	https://www.youtube.com/watch?v=6BJImA0M4Jo
3	www.allaboutcircuits.com
4	http://www.datasheetcatalog.com/
5	https://www.youtube.com/watch